

Compilers' Theory

Lecture 3

Prof. Zaki Taha

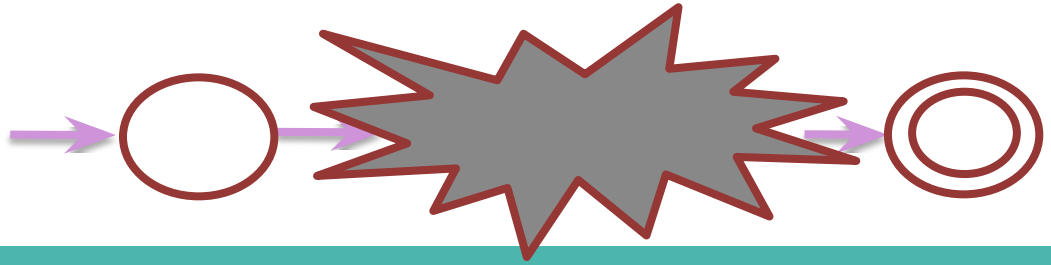
Course Coordinator
Dr. Mohamed Ali
mohamed.abdelghafour@cis.asu.edu.eg



Converting Regular Expression to NFA

Thompson's Construction – Base Rules

- Thompson's Construction translates any Regular Expression (RE) into an equivalent NFA using induction.
- Structural Restrictions:
  - Single Start State: There is exactly one starting state, with no transitions entering it from within the NFA.
  - Single Accepting State: There is exactly one accepting state, with no transitions leaving it.



Regular Expressions

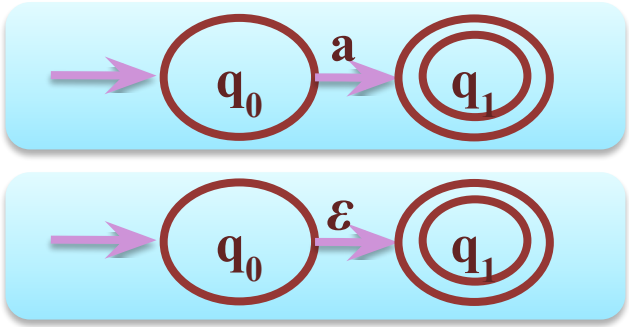
- Single Character
- Or
- Concatenation
- Closure
- Parentheses

Regular Expressions

- Single Character
- Or
- Concatenation
- Closure
- Parentheses

Regular Expressions – Single Character

- A basic regular expression consisting of a single character a, where a is from an alphabet Σ of legal characters, the meta character ε, or the meta character ϕ, where the following holds respectively:
  - $L(a) = \{a\}$
  - $L(\varepsilon) = \{\varepsilon\}$
  - $L(\phi) = \{\ \}$



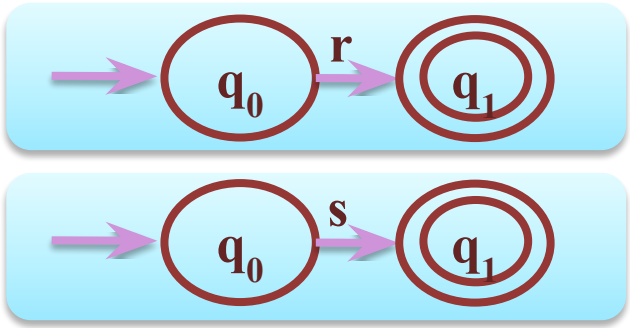


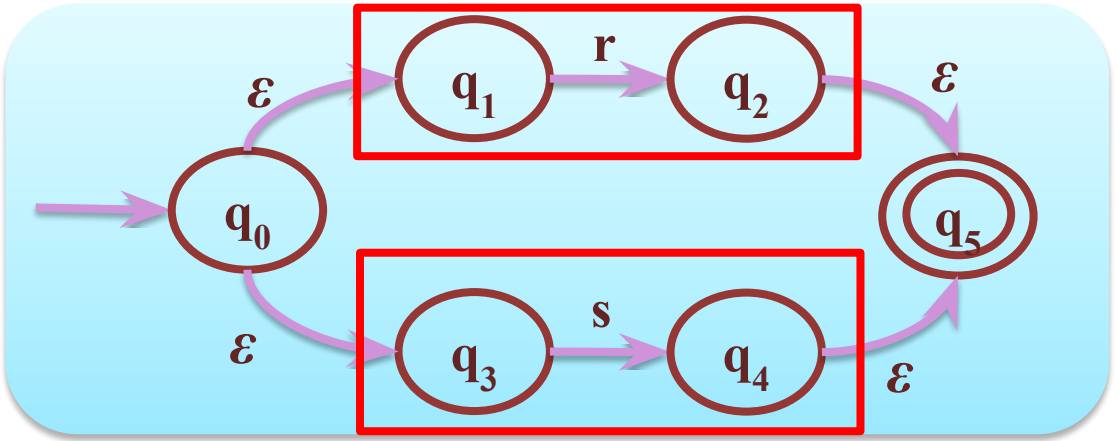
Regular Expressions

Single Character | Or | Concatenation | Closure | Parentheses

Regular Expressions – Or

- An expression of the form ***r*** | ***s***, where *r* and *s* are regular expressions
  - $L(r \mid s) = L(r) \cup L(s)$





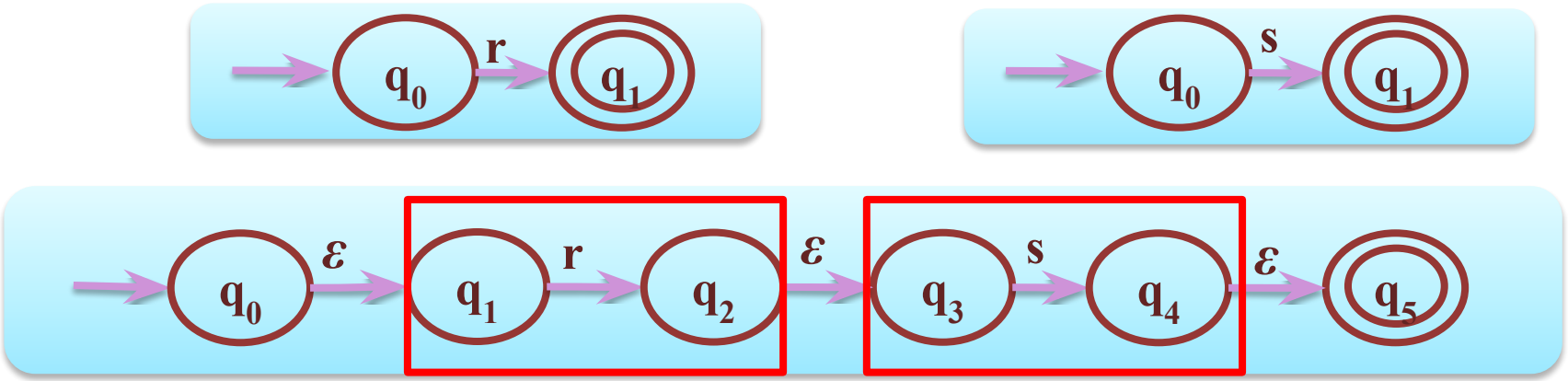


Regular Expressions

Single Character | Or | **Concatenation** | Closure | Parentheses

Regular Expressions – Concatenation

- An expression of the form ***r s***, where *r* and *s* are regular expressions
  - $L(r\,s) = L(r) \,.\, L(s)$



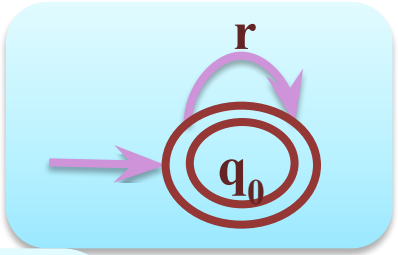


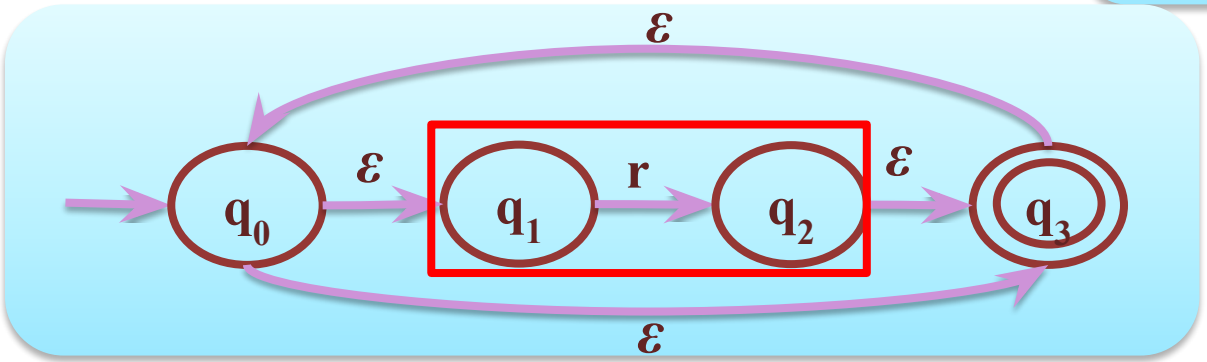
Regular Expressions

Single Character | Or | Concatenation | **Closure** | Parentheses

Regular Expressions – Closure

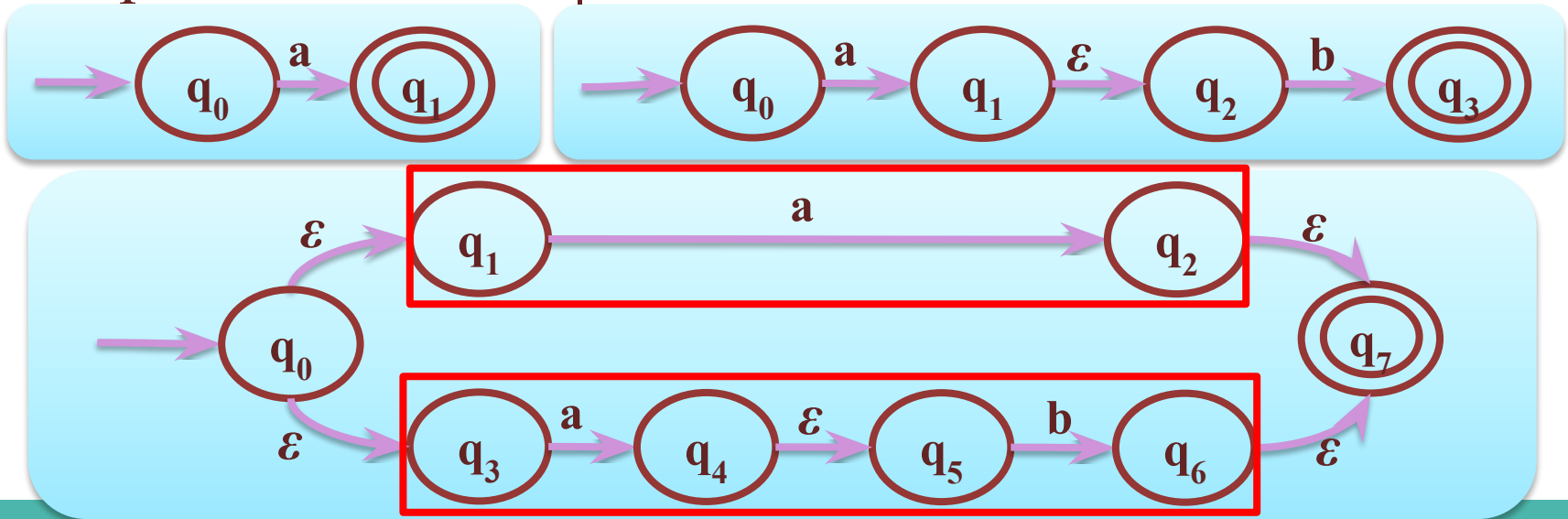
- An expression of the form ***r****, where *r* is a regular expression
  - *L (r*) = L (r)**
  - *L(r*) = {ε, r, rr, rrr, rrrr, ...}*





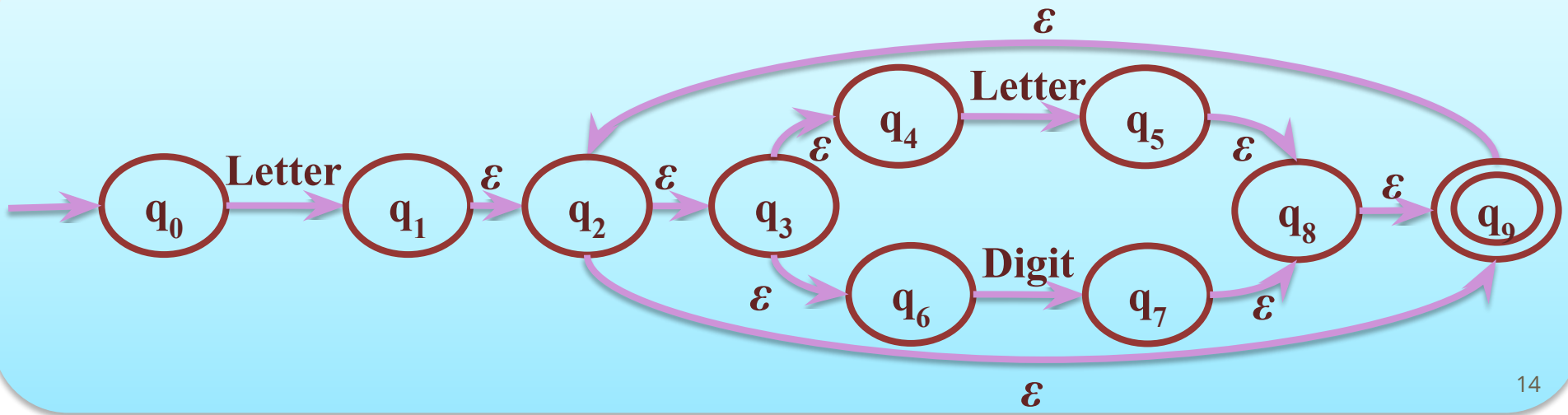
Converting Regular Expression to NFA

1. Construct NFA that accepts the sets of all strings in the language represented by the following regular expression: **ab | a**



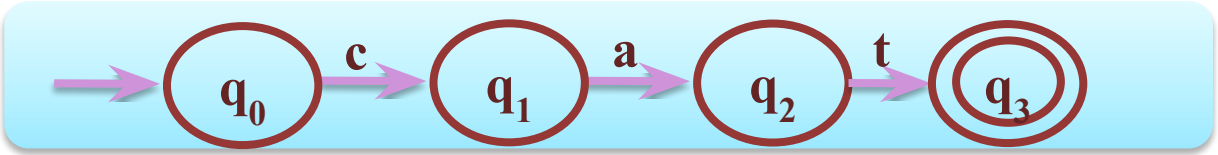
Converting Regular Expression to NFA

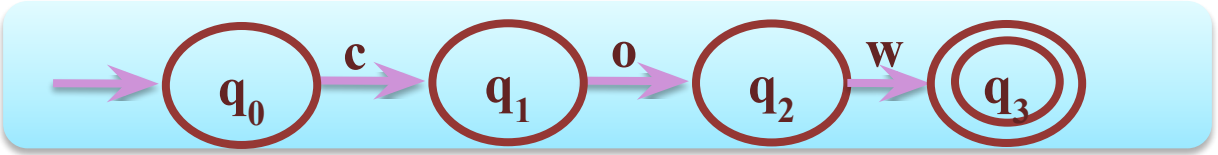
2. Construct NFA that accepts the sets of all strings in the language represented by the following regular expression: **Letter (Letter | Digit)***



Converting Regular Expression to NFA

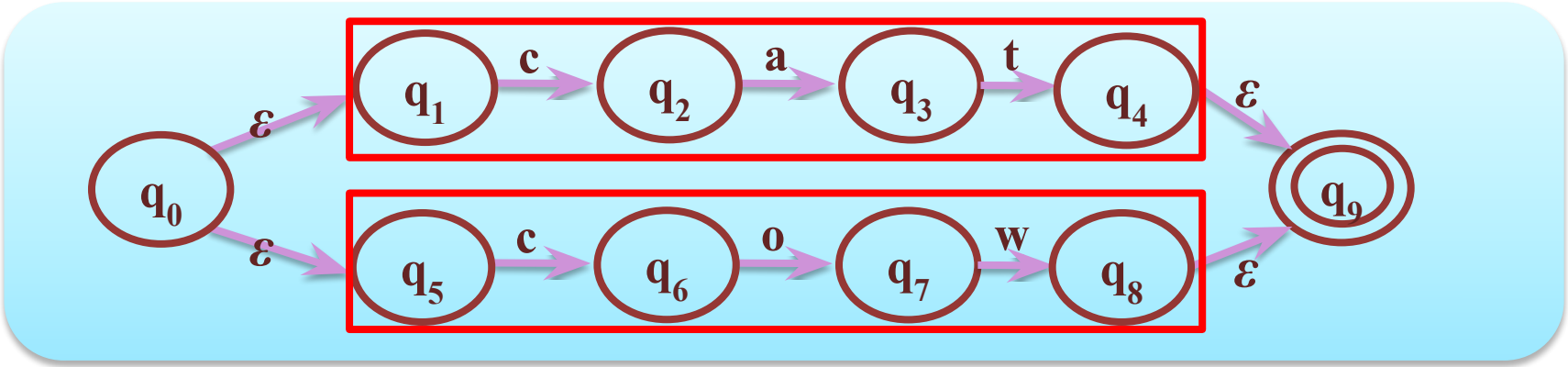
3. Construct NFA that accepts the sets of all strings in the language represented by the following regular expression: **a(cat | cow)***





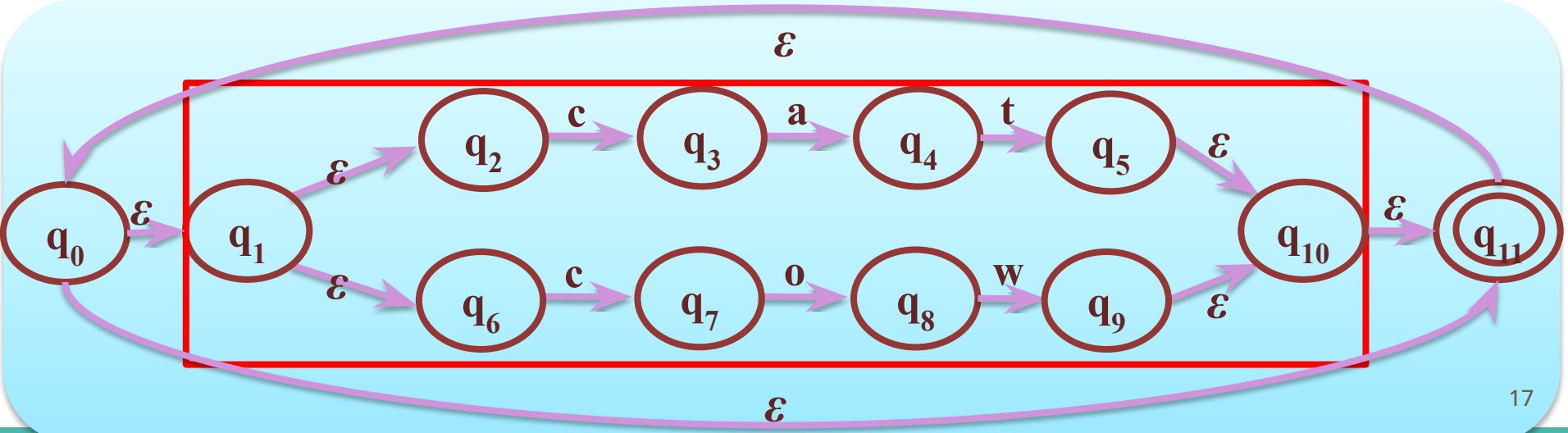
Converting Regular Expression to NFA

3. Construct NFA that accepts the sets of all strings in the language represented by the following regular expression: **a(cat | cow)***

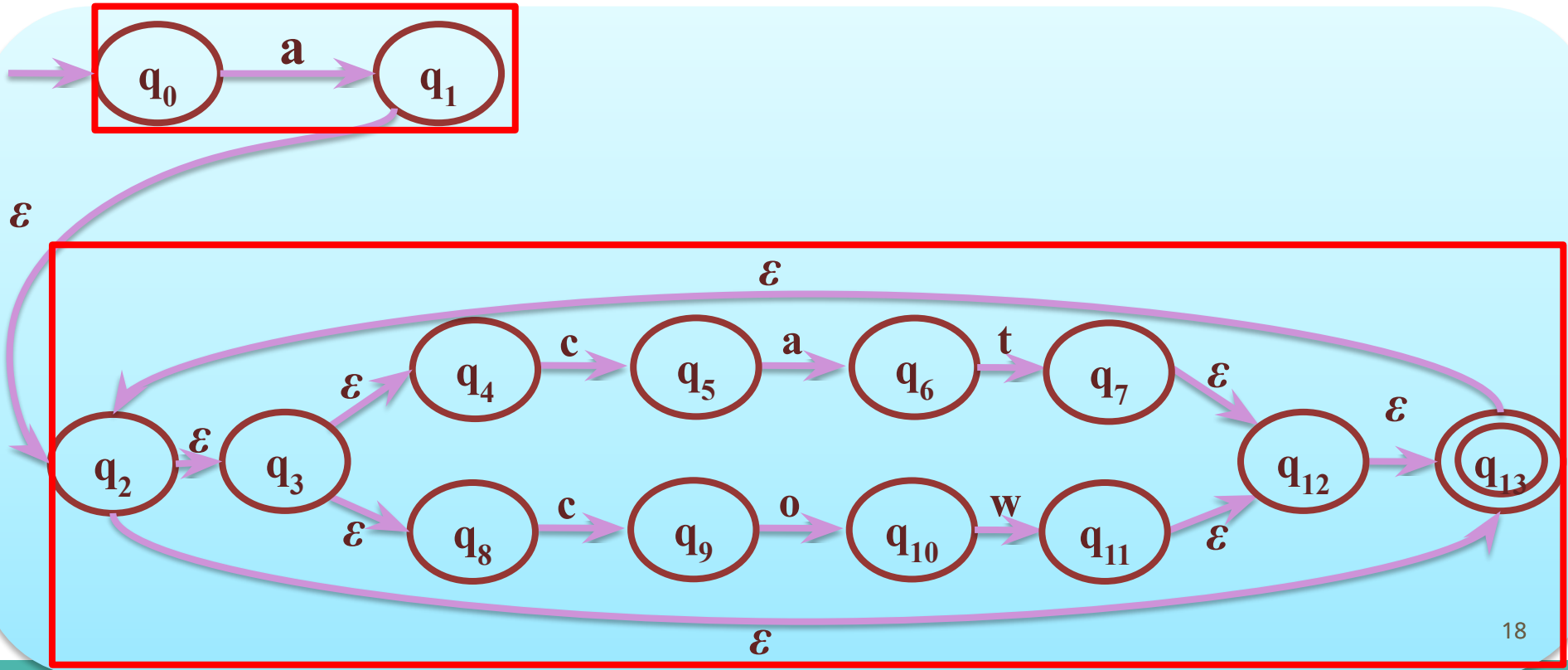


Converting Regular Expression to NFA

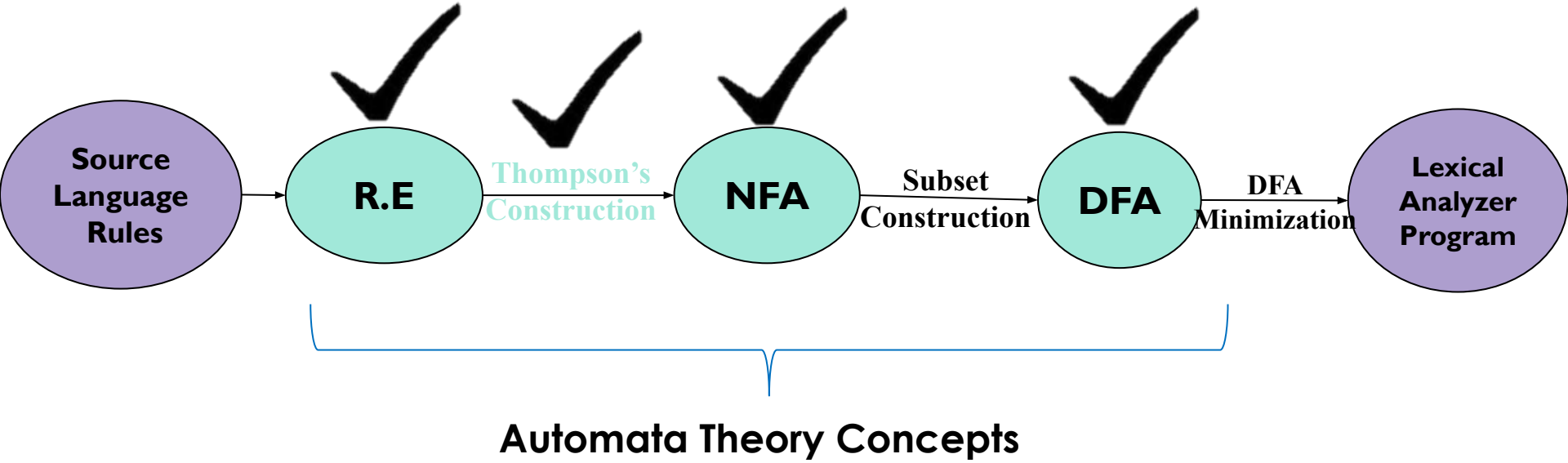
3. Construct NFA that accepts the sets of all strings in the language represented by the following regular expression: **a(cat | cow)***



Converting Regular Expression to NFA



Lexical Analyzer and Automata



SUBSET CONSTRUCTION
NFA -> DFA

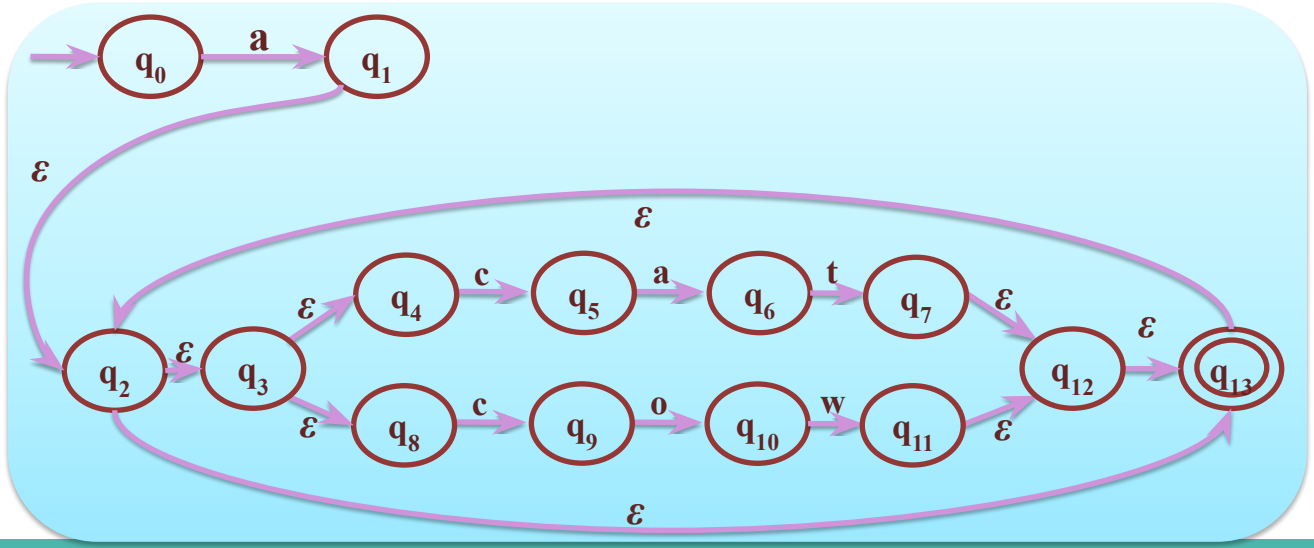
Concept of Subset Construction (NFA -> DFA)



- NFAs are impossible to run directly in constant time; subset construction converts an NFA's active state sets into single, distinct DFA states.
- The Key Function: Epsilon Closure (ε-closure)
  - **ε-closure(n)** is the set of all NFA states reachable from state **n** by traversing only **ε** transitions.

Example (From our a(cat|cow)* NFA)

- **ε-closure(N_0)** = {N_0}
- **ε-closure(N_1)** = {N_1, N_2, N_3, N_4, N_8, N_{13}} (These states are immediately activated after consuming character **a**).

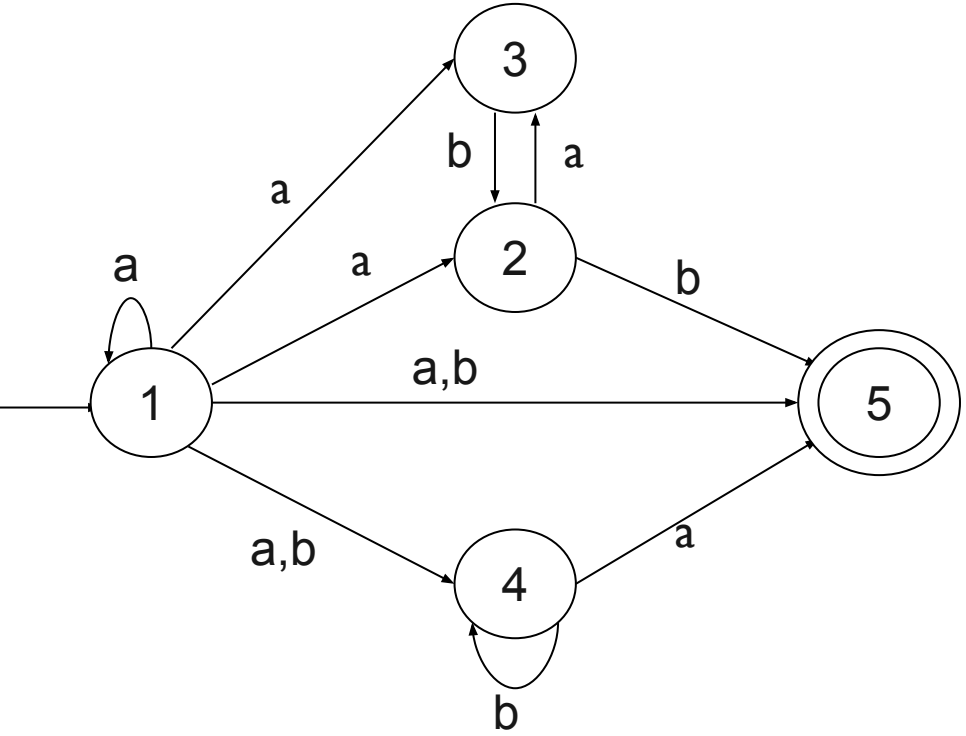


SUBSET CONSTRUCTION METHOD

Using Subset construction method to convert NFA to DFA involves the following steps:

- For every state in the NFA, determine all ***reachable states*** for every input symbol.
- The set of reachable states constitute a ***single state*** in the converted DFA (Each state in the DFA corresponds to a subset of states in the NFA).
- Find ***reachable states*** for each new DFA state, until no more new states can be found.

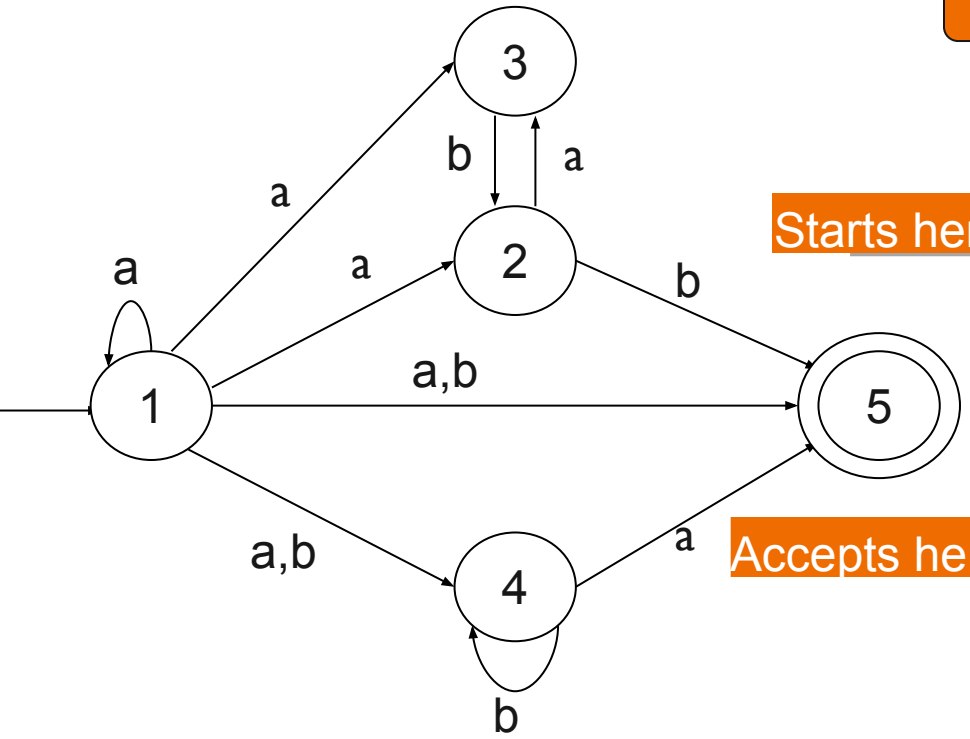
SUBSET CONSTRUCTION METHOD



Step1

Construct a transition table showing all reachable states for every state for every input signal.

SUBSET CONSTRUCTION METHOD



Transition from state q with input a

Transition from state q with input b

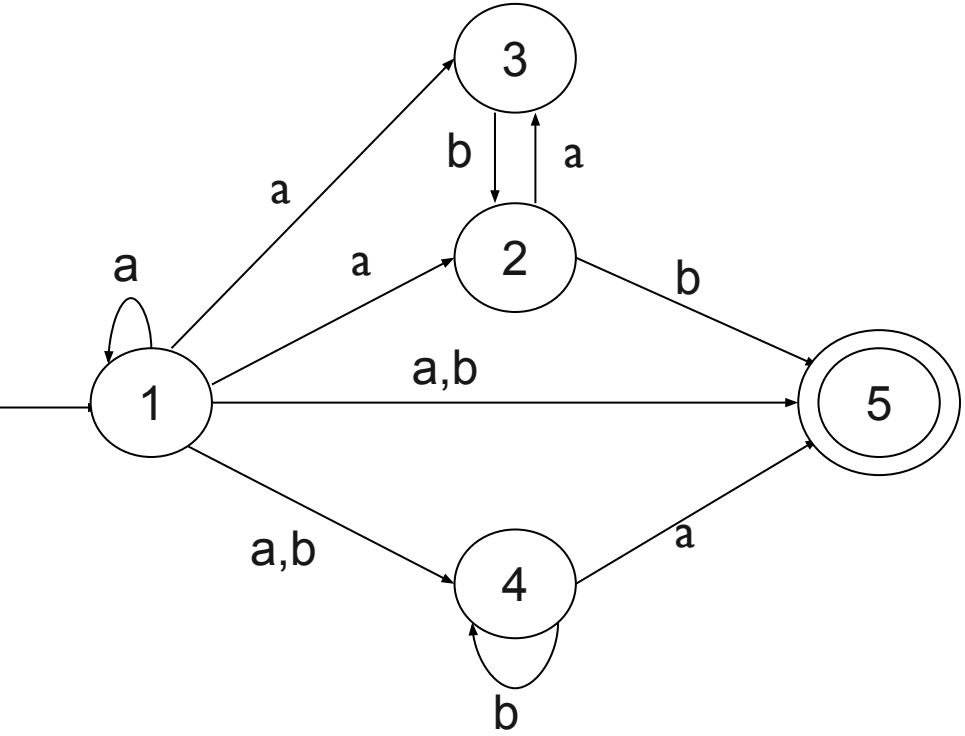
Starts here

Accepts here

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

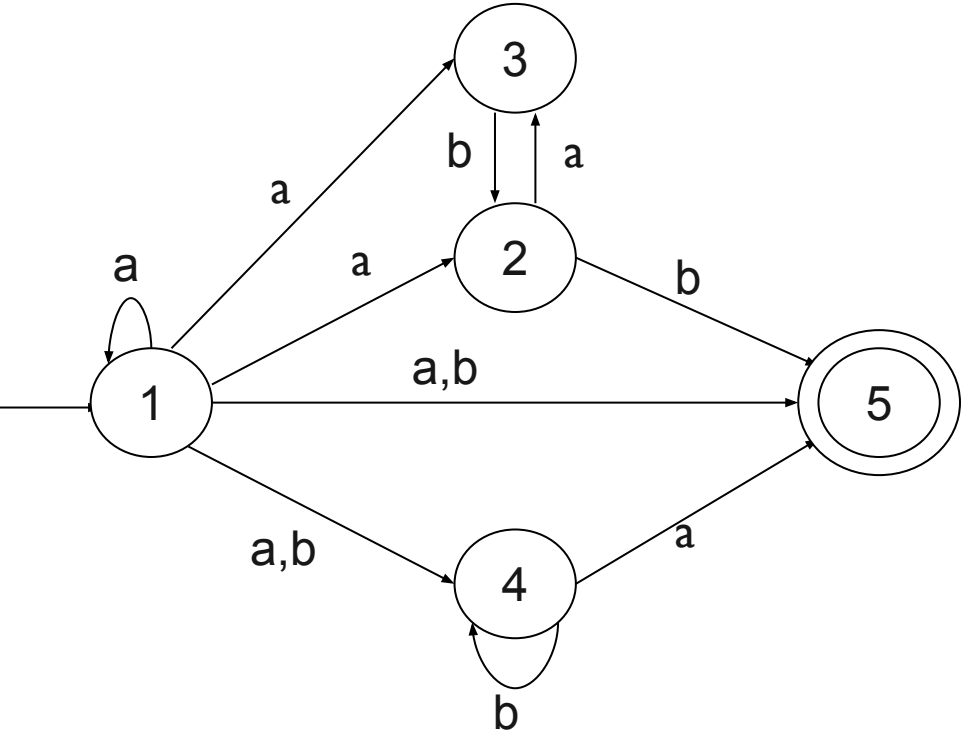
SUBSET CONSTRUCTION METHOD



q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

SUBSET CONSTRUCTION METHOD



Step2

The set of states resulting from every transition function constitutes a new state. Calculate all reachable states for every such state for every input signal.

Starts with Initial state

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

Starts with Initial state

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

Starts with Initial state

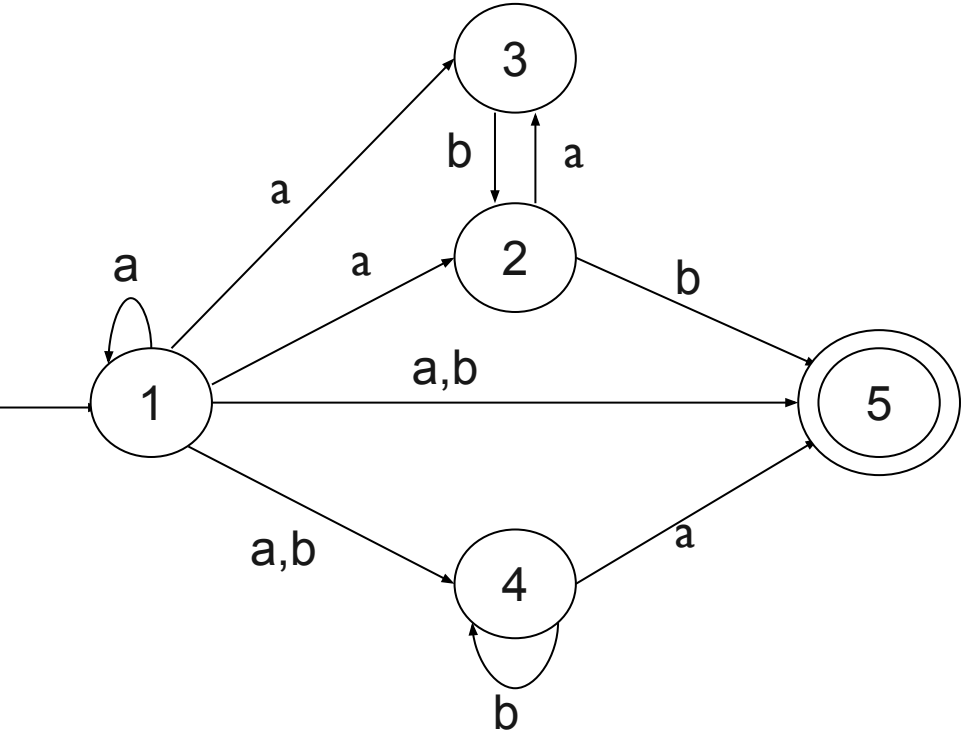
q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*		
{4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

SUBSET CONSTRUCTION METHOD



Step3

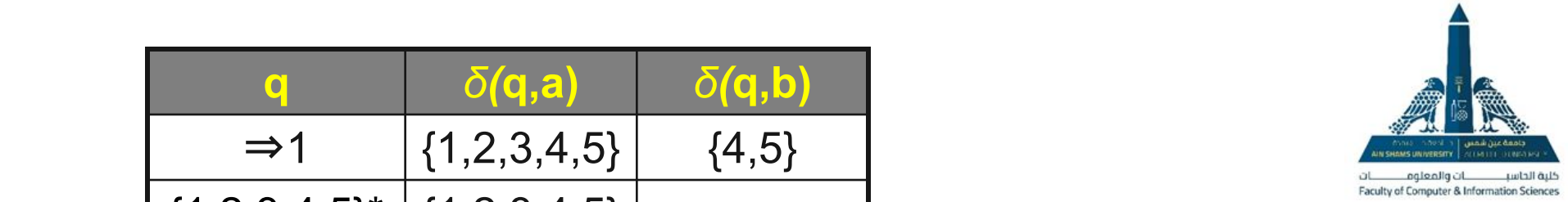
Repeat this process (step2) until no more new states are reachable.

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*		
{4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table



q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	
{4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*		
{2,4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*		
{2,4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	
{2,4,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	
{2,4,5}*		
{5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*		
{5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*		
{5}*		
{4}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*		
{5}*		
{4}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	
{5}*		
{4}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	
{5}*		
{4}		
{3,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*		
{4}		
{3,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*		
{4}		
{3,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	
{4}		
{3,5}*		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	
{4}		
{3,5}*		
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}		
{3,5}*		
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}		
{3,5}*		
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	
{3,5}*		
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*		
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*		
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅		
{2}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅		
{2}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	
{2}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	
{3}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}		

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	{2}

DFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	{2}

DFA Transition table

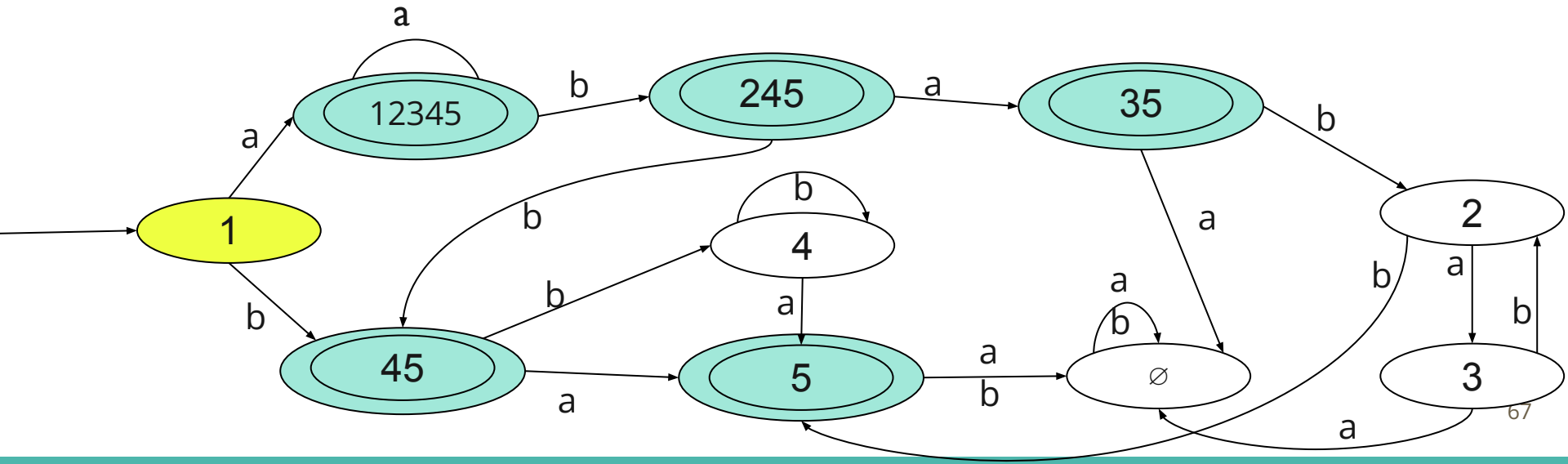
Stops here as there are no more reachable states

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
2	{3}	{5}
3	∅	{2}
4	{5}	{4}
5*	∅	∅

NFA Transition table

q	δ(q,a)	δ(q,b)
⇒1	{1,2,3,4,5}	{4,5}
{1,2,3,4,5}*	{1,2,3,4,5}	{2,4,5}
{4,5}*	{5}	{4}
{2,4,5}*	{3,5}	{4,5}
{5}*	∅	∅

q	δ(q,a)	δ(q,b)
{4}	{5}	{4}
{3,5}*	∅	{2}
∅	∅	∅
{2}	{3}	{5}
{3}	∅	{2}



Lexical Analyzer and Automata

